2025 Exam

$$p(y_1,\theta,z_1) = \left[(1-z_1)\,\mathcal{N}(y_1|0,1) + z_1\,\mathcal{N}(y_1|\theta^T_{x_1},\sigma^2)\right] \mathrm{Ber}(z_1|\tilde{\pi})\,\mathcal{N}(\theta|0,\alpha^{-1}I)$$

3.1 Determine the distribution $p(y_1,\theta)$ analytically

$$p(y_1,\theta,z_1) = P(y_1|\theta,z_1)\,P(\theta)\,P(z_1)$$

a) $z=0$

$$\left[(1-0)\,\mathcal{N}(y_1|0,1) + 0\cdot\mathcal{N}(y_1|\theta^T_{x_1},\sigma^2)\right](1-\tilde{\pi})\,\mathcal{N}(\theta|0,\alpha^{-1})$$

$$= \left[\mathcal{N}(y_1|0,1)\right](1-\tilde{\pi})\,\mathcal{N}(\theta|0,\alpha^{-1}I)$$

a) $z=1$

$$\left[(1-1)\,\mathcal{N}(y_1|0,1) + 1\cdot\mathcal{N}(y_1|\theta^T_{x_1},\sigma^2)\right]$$

$$= \left[\mathcal{N}(y_1|\theta^T_{x_1},\sigma^2)\right]\tilde{\pi}\,\mathcal{N}(\theta|0,\alpha^{-1}I)$$

$$p(y,\theta) = \sum_{z_1\{0,1\}} p(y_1,\theta,z_1) = \left[\mathcal{N}(y_1|0,1)\right](1-\tilde{\pi})\,\mathcal{N}(\theta|0,\alpha^{-1}I) + \left[\mathcal{N}(y_1|\theta^T_{x_1},\sigma^2)\right]\tilde{\pi}\,\mathcal{N}(\theta|0,\alpha^{-1}I)$$

3.2 Determine the distribution $p(y_1,z)$ analytically

$$p(y_1,\theta,z_1) = P(y_1|\theta,z_1)\,P(\theta)\,P(z_1)$$

$$p(y_1,z_1) = \int \left[(1-z_1)\,\mathcal{N}(y_1|0,1) + z_1\,\mathcal{N}(y_1|\theta^T_{x_1},\sigma^2)\right]\mathrm{Ber}(z_1|\tilde{\pi})\,\mathcal{N}(\theta|0,\alpha^{-1}I)\,d\theta$$

$$= \int \left[(1-z_1)\,\mathcal{N}(y_1|0,1)\,\mathcal{N}(\theta|0,\alpha^{-1}I) + z_1\,\mathcal{N}(y_1|\theta^T_{x_1},\sigma^2)\,\mathcal{N}(\theta|0,\alpha^{-1}I\right]\mathrm{Ber}(z_1|\tilde{\pi})\,d\theta$$

$$= \left[(1-z_1)\,\mathcal{N}(y_1|0,1)\underbrace{\int\mathcal{N}(\theta|0,\alpha^{-1}I)}_{1} + z_1\,\mathcal{N}(y_1|\theta^T_{x_1},\sigma^2)\,\mathcal{N}(\theta|0\,\alpha^{-1}I)\,d\theta\right]\mathrm{Ber}(z_1|\tilde{\pi})$$

$$= \left[(1-z_1)\,\mathcal{N}(y_1|0,1)\cdot 1 + z_1\cdot\mathcal{N}(y_1|0,\sigma^2+x\,\alpha^{-1}I\,x^T\right]\mathrm{Ber}(z_1|\tilde{\pi})$$

$$p(y) = \sum_{z_1\in\{0,1\}} p(y_1,z_1)$$

a) $z=0$

$$\left[(1-0)\,\mathcal{N}(y_1|0,1)\,(1-\tilde{\pi}) + 0\cdot\mathcal{N}(y_1|0,\sigma^2+X\alpha^{-1}X^T\right]$$

b) $z=1$

$$\left[(1-1)\,\mathcal{N}(y_1|0,1) + 1\cdot\mathcal{N}(y_1|0,\sigma^2+X\alpha^{-1}X^T)\cdot\tilde{\pi}\right.$$

$$P(y) = \sum_{z_1 \in \{0,1\}} p(y_1, z_1)$$

$$= \underbrace{\mathcal{N}(y_1 \mid 0, 1)\,(1-\pi)}_{P(z=0)} + \underbrace{\mathcal{N}(y_1 \mid 0, \sigma^2 + X\alpha^{-1}X^T)\,\pi}_{P(z=1 \mid y)}$$

- we are given single data pt. (x_1, y_1) w/ $x_1 \in \mathbb{R}^d$

model: $z_1 \sim Ber(\pi)$
$\theta \sim \mathcal{N}(0, \lambda^{-1} I)$

Conditional likelihood $y_1 | \theta, z_1 \sim \begin{cases} \mathcal{N}(0, \tau^2) & , z_1 = 0 \\ \mathcal{N}(\theta^T x_1, \sigma^2) & z_1 = 1 \end{cases}$

Questions: 1. Write the joint $p(y_1, z_1, \theta)$

$$p(y_1, z_1, \theta) = \left[(1-z_1)\mathcal{N}(y|0, \tau^2) + z_1 \mathcal{N}(y|\theta^T x_1, \sigma^2)\right] Ber(z_1|\pi)\,\mathcal{N}(0, \lambda^{-1} I)$$

2. Marginalize z_1 to get $p(y_1, \theta)$

$$p(y_1, \theta) = \sum_{z_1 \in \{0,1\}} p(y_1|\theta, z_1)\, p(z_1)\, p(\theta)$$

a) sum over $z_1 = 0$

$$\left[(1-0)\mathcal{N}(y|0, \tau^2) + 0\,\mathcal{N}(y|\theta^T x_1, \sigma^2)\right](1-\pi)\,\mathcal{N}(0, \lambda^{-1} I)$$
$$\left[\mathcal{N}(y|0, \tau^2)\right](1-\pi)\,\mathcal{N}(0, \lambda^{-1} I)$$

sum over $z_1 = 1$

$$\left[(1-1)\mathcal{N}(y|0, \tau^2) + 1\,\mathcal{N}(y|\theta^T x_1, \sigma^2)\right] p\,\mathcal{N}(y|0, \lambda^{-1} I)$$
$$\mathcal{N}(y|\theta^T x_1, \sigma^2)\, p\,(\mathcal{N}(y|0, \lambda^{-1} I))$$

$$p(y_1, \theta) = \sum_{z_1 \in \{0,1\}} p(y_1|\theta, z_1)\, p(z_1)\, p(\theta)$$
$$= \left[\mathcal{N}(y|0, \tau^2)(1-\pi) + \mathcal{N}(y|\theta^T x_1, \sigma^2)\,\pi\right] p\,(\mathcal{N}(y|0, \lambda^{-1} I))$$

3. Marginalize θ to get $p(y_1, z_1)$